Data Professional Survey Analysis Dashboard

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1. Problem Statement

The data industry has grown rapidly over the past decade, yet comprehensive, structured intelligence about what it is like to work in the field remains scarce. Aspiring data professionals, hiring managers, and HR teams frequently make decisions — on salary benchmarks, tool investments, or upskilling paths — based on anecdotal evidence rather than surveyed data.

This project addresses the following core challenges:

- Salary opacity – there is little standardised, publicly available data on what data professionals earn across different roles, regions, and industries.
- Unquantified satisfaction – subjective metrics like work-life balance and management quality are rarely captured in a structured, comparable format.
- Technology fragmentation – the data tooling landscape is vast; understanding which programming languages professionals rely on informs curricula and hiring criteria.

The objective of this project is to analyse the Data Professional Survey dataset (630 respondents, 28 variables, collected June 2022) and build an interactive Power BI dashboard that surfaces insights into role distribution, salary patterns, language preferences, satisfaction scores, entry difficulty, and global geographic spread.

2. Solution Overview

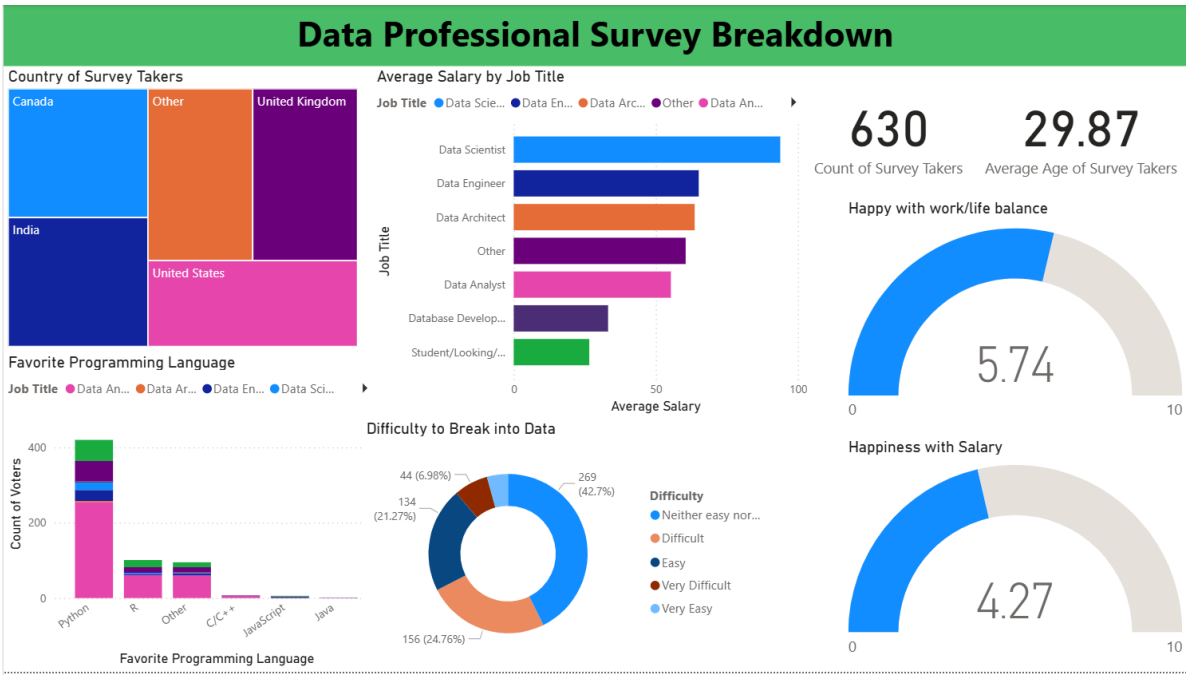
The solution is a multi-page Power BI dashboard that transforms 630 raw survey responses into a structured, interactive analytical experience. The end-to-end pipeline covers data ingestion, cleaning, modelling, and visualization.

2.1 Data Pipeline

- Source data – a single Excel workbook (630 rows × 28 columns) containing anonymised survey responses.

- Power Query cleaning – removed incomplete rows, standardised salary bracket labels, normalised free-text role fields, and split compound entries in the programming language column.
- Calculated measures via DAX – average salary by role, average happiness per dimension, career-switch percentage, and total respondent count.

2.2 Dashboard Visualization



Visualization	Purpose	Key Insight
KPI Cards	High-Level Snapshot	Displays total survey count, average age (29.87), gender split (74.3% Male / 25.7% Female), and average salary satisfaction (4.27/10).

Bar Chart – Avg Salary by Role	Role vs Earnings	Data Scientists and Engineers lead in average salary; Data Analysts cluster in the 41k–85k band.
Bar Chart – Favourite Language	Technology Preferences	Python dominates at 66.7% of respondents; R follows at 16%, with SQL and others trailing.
Donut Chart – Career Switchers	Career Transition	59% of respondents switched into data from another field, reflecting the industry's cross-disciplinary appeal.
Tree map – Respondents by Country	Geographic Spread	US (261), India (73), UK (40), Canada (32), and Nigeria (18) are the top five represented nations.
Gauge – Happiness: Salary	Satisfaction Score	Average score of 4.27/10 — the lowest of all happiness metrics — signalling persistent salary dissatisfaction.
Gauge – Happiness: Work/Life	Satisfaction Score	Average score of 5.74/10 — the highest happiness metric — showing relatively positive work-life balance sentiment.
Stacked Bar – Entry Difficulty	Barrier to Entry	42.7% rated it 'Neither easy nor difficult'; combined 'Difficult/Very Difficult' responses reached 31.7%.
Country Slicer	Dynamic Filtering	Allows filtering all visuals by selected country, enabling instant drill-down into regional professional trends.

3. Key Findings

Analysis of the 630-response dataset revealed eight substantive findings that paint a clear portrait of the current data profession landscape:

Key Finding	Detail
Python Dominance	420 of 630 respondents (66.7%) prefer Python, far ahead of R (16%) and SQL variants (~6%).

Salary Dissatisfaction	Average salary happiness of 4.27/10 is the lowest across all six measured dimensions.
Career Switching is Common	59% of respondents transitioned into data from a different career, demonstrating the field's accessibility.
US Leads Globally	The United States accounts for 41.4% of all respondents (261/630), followed by India (11.6%).
Moderate Entry Difficulty	Only 42.7% found entry 'Neither easy nor difficult'; 31.7% found it Difficult or Very Difficult.
Young Professional Base	Average respondent age is 29.87, suggesting data careers attract a predominantly early-career workforce.
Education Skews High	52.2% hold a bachelor's degree; 30.5% hold a Master's; only 0.8% have a PhD.
Remote Work is Top Priority	When asked about new-job priorities, Remote Work and Better Salary ranked highest among preferences.

3.1 Happiness Score Breakdown

Respondents rated their satisfaction across six workplace dimensions on a 0–10 scale. The results reveal a consistent pattern: interpersonal and developmental satisfaction is higher than economic and advancement satisfaction.

Dimension	Avg Score	Trend	Interpretation
Salary	4.27 / 10	⬇ Lowest	Strong signal of underpayment, especially at junior levels.
Work / Life Balance	5.74 / 10	⬆ Highest	Relatively positive; remote and hybrid roles likely a factor.
Coworkers	5.86 / 10	⬆ High	Collaborative data teams foster good peer relationships.
Management	5.33 / 10	➡ Mid	Moderate satisfaction: leadership quality varies by company.
Upward Mobility	4.76 / 10	⬇ Low	Career advancement pathways still unclear in many organisations.

Learning New Things	5.61 / 10	⬆ High	Data professionals value continuous skill development highly.

4. Tech Stack

The following tools and technologies were used to design, develop, and publish this project:

Tool	Role	Usage in This Project
Power BI	Dashboard & Visualization	KPI cards, bar/donut/tree map charts, gauge visuals, and interactive slicers for role and country filtering.
Power Query	Data Transformation	Cleaned raw survey data, standardized salary brackets, renamed columns, removed nulls, and split multi-value fields.
DAX	Calculated Measures	Measures for average salary by role, average happiness scores, career-switch %, and respondent counts.
Excel (.xlsx)	Source Data	630-row survey dataset with 28 columns including role, salary, language preference, happiness scores, and demographics.

5. Unique Features

- Role-wise salary benchmarking – Rather than showing a single average salary, the dashboard breaks down earnings by job title (Data Analyst, Data Scientist, Data Engineer, etc.), enabling targeted salary comparisons that are immediately useful to both job seekers and recruiters.

- Six-dimension happiness heatmap – Instead of a single satisfaction score, the dashboard presents all six happiness dimensions side-by-side as gauge visuals and a summary table, making it possible to identify the specific pain

points (salary, upward mobility) versus the strengths (coworkers, work-life balance).

- Career switch vs. native entrant lens – A dedicated visual segments all metrics by whether the respondent switched careers into data, revealing whether career-switchers earn more, are happier, or find entry harder than native entrants.

- Country-level tree map and slicer – A tree map visualizes global participation by country size, and a slicer propagates country-based filters across all visuals simultaneously, enabling region-specific analysis without building separate dashboards.

- Clean, layered dashboard architecture – KPI cards anchor the top tier; bar and donut charts occupy the analytical mid-tier; gauge visuals provide nuanced satisfaction scoring at the bottom — creating a clear top-down reading flow from summary to detail.

6. Future Improvements

- Annual longitudinal tracking – Repeat the survey annually and build a year-over-year comparison view to track how salary satisfaction, language preferences, and entry difficulty evolve over time as the data industry matures.
- Industry-level salary segmentation – Break down salary not just by role but by the industry the respondent works in (Finance, Healthcare, Tech, etc.) to surface sector-specific compensation benchmarks.
- Machine learning integration via Python – Use Power BI's Python visuals to run a clustering model (e.g., K-Means) on happiness dimensions and salary to identify distinct professional archetypes — for example, well-paid but unhappy vs. modestly paid but satisfied.
- Predictive salary estimator – Build a DAX or Python-powered tool within the dashboard where a user selects their role, country, and years of

experience, and the model returns an estimated salary band based on survey data.

- Mobile-optimised layout – Design a dedicated phone-layout view in Power BI Mobile to ensure the dashboard is accessible and readable on smaller screens without needing to zoom or scroll excessively.
- Expanded demographic analysis – The current dashboard gives limited visibility into how education level, ethnicity, and age interact with salary and satisfaction. Future iterations should add multi-dimensional demographic breakdowns to surface equity and diversity insights.

7. Conclusion

This project successfully demonstrates how Power BI — combined with Power Query for data preparation and DAX for analytical measures — can convert a raw 630-row survey export into a rich, multi-dimensional professional insights dashboard.

The dashboard surfaces findings that are both practically valuable and analytically nuanced: Python's dominance as the tool of choice, the persistent salary dissatisfaction that contrasts with high coworker and work-life satisfaction, the prevalence of career switchers, and the genuinely global nature of data as a profession. These are not insights visible in the raw spreadsheet; they emerge only through thoughtful visualization.

Beyond the technical execution, this project demonstrates the importance of asking the right questions of data. The dataset could have been summarised in a few averages. Instead, by structuring the analysis around professional experience, satisfaction dimensions, and geographic context, it becomes a resource that speaks directly to the lived reality of data professionals worldwide.